PRODUCT & UI BLUEPRINT

Station Admin Dashboard

IoT washroom access management for petrol stations

A focused operations workspace for one petrol station

The station administrator can monitor card access, manage people and credentials, review access history, maintain schedules, and respond to device or door events. The company Super Admin retains cross-station oversight and platform-level control.

9	12	2	1
Sidebar pages	Overview components	User roles	Station scope

Prepared from the supplied DHI-ASI1201E / SmartPSS Lite feature summary. Confirmed functions are separated from documented capabilities that still require firmware, wiring and client-version validation.

Design principle: show operationally useful information first; do not fill the screen with decorative charts or metrics the controller cannot actually provide.

1. Product scope and role boundaries

The system has two roles: a Station Admin assigned to one petrol station, and a company Super Admin who manages the overall platform and multiple stations. This document specifies the Station Admin dashboard only.

Station Admin responsibilities

• Manage people and their card credentials for the assigned station.

• Set credential validity and assign available door/time-section permissions.

• Monitor the assigned controller's online state and door/access events where supported.

• Review successful and denied access records and investigate unusual activity.

• View station-level usage and export operational reports.

Super Admin boundary (not part of this station dashboard)

The company Super Admin should have a separate company console for station creation, station-admin accounts, cross-station analytics, device fleet oversight, global configuration and subscription/commercial controls. A Station Admin should not see other stations or company-wide settings.

2. Recommended sidebar — 9 pages

1	Overview	Station command centre	Today’s scans, success/denial, device status, trends, recent e
2	Live Monitoring	Near-real-time operational view	Latest scans, controller connectivity, door state if sensor exist
3	Access Logs	Searchable audit history	Timestamp, card/person, result, reason/event, door/device
4	People & Cards	Manage users and credentials	Person profile, card number, category/status, validity
5	Access Schedules	Manage when credentials can enter	Time sections, weekdays, holidays, assigned permissions
6	Device & Door	Monitor the station controller	Device identity, online status, heartbeat if integrated, relay/doo
7	Alerts & Events	Investigate exceptions	Door timeout, intrusion, tamper, duress, illegal/blocklist card a
8	Reports	Review and export station performance	Daily/weekly/monthly access summaries, outcomes, peak per
9	Settings	Station-level configuration	Station profile, local preferences, notification rules, admin prof

Navigation note: keep the sidebar fixed and highlight the active page. Put the station name and a station switcher only if the same admin is allowed to manage more than one assigned station. Otherwise show the station name as a fixed context label.

3. Page 1 — Overview (the most detailed page)

This is the landing page and should provide a complete operational snapshot without requiring the admin to open other pages. Recommended layout: a compact header, KPI strip, two chart rows, then live activity and device/event panels.

1. Page header + date filter	Station name, current date, last updated time; Today / 7 days / 30 days / custom range	All charts and summary cards respond to the selected range.
2. System status banner	Overall status: Operational / Attention required / Offline	Derived from controller connectivity and active critical alerts. Avoid claiming full system health if only one signal is available.
3. KPI — Total scans	Number of card authentication attempts in selected range	Count all recorded attempts, including denied attempts.
4. KPI — Granted access	Successful card authentications	Count successful grants; show comparison to previous equivalent period.
5. KPI — Denied access	Rejected/unauthorized attempts	Count denied attempts; allow click-through to filtered Access Logs.
6. KPI — Success rate	Granted ÷ total attempts × 100	Show denominator and date range; do not use as a device uptime metric.
7. KPI — Controller status	Online / Offline based on controller connection	Use SmartPSS/client connection status or the integration's own verified heartbeat.
8. Access activity line chart	Granted and denied scans over time (hourly for today; daily for longer ranges)	Two series; hover tooltip with count; click point to open matching logs.
9. Access outcome donut	Granted vs denied share	Two segments; show count and percentage; use same selected date range.
10. Usage by hour bar chart	Scan volume grouped by hour	Highlights peak washroom usage; only show if timestamps are reliably stored.
11. Recent access activity table	Latest 5–10 attempts	Time, masked card ID, person (if resolved), result, reason if available.
12. Device & door status panel	Controller online status and door state where a door contact is installed	Show last seen/heartbeat only if collected. Door open/closed requires a functioning door sensor; relay command alone is not proof of physical door state.

Recommended visual arrangement

Row 1	4 KPI cards: Total scans | Granted | Denied | Success rate
Row 2	Wide access activity line chart (about 2/3 width) + outcome donut (about 1/3 width)
Row 3	Usage-by-hour bar chart (about 1/2 width) + device/door status panel (about 1/2 width)
Bottom	Recent activity table with "View all logs" link

Avoid adding a calendar to the overview just to make it look busy. A calendar belongs in a future maintenance workflow only if the business will actually schedule inspections, service visits or planned downtime.

4. Page 2 — Live Monitoring

Purpose: a live operations screen for the station admin, not a second analytics dashboard.

1. Live status header	Controller connection state, last communication time, live-feed state	Show “Live” only if updates are genuinely pushed or polled at a defined interval.
2. Latest scan feed	Chronological stream of card attempts	Time, masked card ID, person, granted/denied, station door/device.
3. Door status tile	Open / closed / unknown	Only show actual physical door state if a door contact sensor is installed and integrated. Otherwise show lock relay command state.
4. Controller status tile	Online/offline and device identifier	Connection state must come from the connected controller/client.
5. Event filters	All / granted / denied / alarms	Filter feed without leaving the page.
6. Quick actions	Open Access Logs; open Device & Door	No remote unlock button by default. If added, require permission, confirmation and audit logging.

5. Page 3 — Access Logs

Purpose: a reliable, searchable audit trail of every recorded authentication attempt.

1. Search	Card number, person name or event ID	Mask card identifiers in general views; reveal full value only to authorized roles.
2. Date/time range	Start and end date/time	Default to today; support custom range.
3. Result filter	Granted / denied / all	Allow quick filtering from overview KPI cards.
4. Person/card filter	Person, card or category	Use only fields available in stored event data.
5. Device/event filter	Controller, door or event type	Useful when a station has multiple controllers/readers.
6. Logs table	Timestamp, person, masked card, result, event/reason, device	Paginate or virtualize large histories; sort newest first.
7. Record detail drawer	Full event details and linked person/card	Display raw controller event code only if available and mapped.
8. Export action	CSV/PDF export for selected range	Apply the same filters to exported data.

6. Page 4 — People & Cards

Purpose: manage the people records and credentials that can authenticate at the station.

1. Summary strip	Total people, active cards, expired cards, blocked cards	Counts derived from the station’s person/card records.
2. People search + filters	Name, card number, category, status	Filter Normal, VIP, Guest, Patrol or Blocklist only where exposed by firmware/client.
3. People table	Name/person ID, user category, status, linked card, validity	Show status and expiry clearly.
4. Add/edit person form	Person details and supported user type/status	Keep fields aligned with the SmartPSS/device integration.
5. Card enrollment flow	Read/register IC card and associate it with a person	Tested setup used a 13.56 MHz MIFARE card.

6. Credential validity editor	Start date and end date	The tested setup exposed credential validity dates.
7. Card actions	Activate/deactivate/block, replace or unlink card	Only expose actions supported by the integration and audit every change.
8. Person detail drawer	Profile, assigned credentials, validity and access permissions	Show credential history if your application stores it.

7. Page 5 — Access Schedules

Purpose: control when a person/card is allowed to use the washroom.

1. Schedule list	Schedule name, time ranges, days and status	Represent device time-section groups in a human-readable way.
2. Weekly schedule editor	Days of week and time intervals	Validate overlaps and empty schedules.
3. Holiday calendar	Holiday dates and special access periods	Dahua documentation lists holiday periods; validate actual behavior on the installed unit.
4. Door/permission assignment	Which door/time section is assigned to a credential	The tested records showed schedule and door-permission fields.
5. Credential validity summary	Start/end validity date for the selected person/card	Keep separate from daily time schedule.
6. Preview/test panel	“This card can access at…” preview	Preview must use the same rules as the controller; otherwise label it as an estimate.
7. Save/publish action	Send updated schedule/permission to controller	Show sync success/failure and last sync time.

8. Page 6 — Device & Door

Purpose: help the station admin understand whether the access-control hardware is connected and operating.

1. Device summary card	Device name/ID, model, online status, station assignment	Model in the supplied setup: Dahua DHI-ASI1201E.
2. Connection details	IP/connection method, client connection status, last seen	Ethernet/TCP-IP was used in the tested setup. Store credentials securely; never display passwords.
3. Door/relay status	Relay command/status; physical door contact state if wired	The controller has one lock relay output and one door-sensor input per the supplied summary. Physical sensor operation needs verification.
4. Device configuration	Door name, device label and supported integration settings	Do not expose unsupported low-level settings to station admins.
5. Sync status	Last successful config sync, pending changes, sync errors	Requires your application to track configuration sync.
6. Device event history	Online/offline changes, restarts and controller events	Only show events actually captured by the integration.
7. Troubleshooting panel	Connection checklist and last error	Provide safe guidance; do not imply remote repair if unavailable.

9. Page 7 — Alerts & Events

Purpose: surface exceptions that may need attention. Alerts depend on supported hardware, wiring, firmware and event integration.

1. Alert summary	Active, acknowledged and resolved counts	Counts from the alert/event service.
2. Severity filters	Critical / warning / info	Define severity mapping centrally.
3. Alert list	Time, event, device, severity, status	Possible documented events include door timeout, intrusion, duress and tamper.
4. Alert detail	Event explanation, related device/person if available	Do not invent a person identity if the event only identifies a card or event code.
5. Acknowledge/resolve	Admin acknowledgement and resolution note	Store actor, timestamp and note in your application audit log.
6. Notification preferences	In-app/email notification settings	Only show channels actually implemented.

10. Page 8 — Reports

Purpose: let the station admin understand usage patterns and export records.

1. Report date range	Today, 7 days, 30 days, custom	All report blocks use the same date range.
2. Usage trend chart	Daily scan totals and granted/denied split	Avoid double-counting retries if event IDs are available.
3. Hourly usage chart	Scan distribution by hour	Useful for staffing/cleaning planning, not a direct occupancy measure.
4. Outcome breakdown	Granted vs denied totals and percentages	Show exact counts alongside percentages.
5. Top usage periods	Busiest hours/days	Derived from stored scan timestamps.
6. Denial reasons table	Most frequent rejection/event reasons	Only if the controller/integration provides reason codes.
7. Export controls	CSV/PDF and selected report type	Respect filters and station permissions.

11. Page 9 — Settings

Purpose: station-level preferences and the station administrator’s own account settings.

1. Station profile	Station name, address/branch code and contact details	Read-only if managed centrally by Super Admin.
2. Admin profile	Name, email, password/security settings	Do not allow station admin to elevate their own role.
3. Notification settings	Which supported alerts trigger notifications	Keep defaults centrally manageable.
4. Display preferences	Timezone, date format and table density	Timezone should match the station location.
5. Security/session settings	Active sessions and sign-out options	Only include if authentication system supports it.
6. Integration information	Controller model, integration state and support reference	Hide secrets, API keys and device passwords.

12. Data model — what the dashboard needs to store

The dashboard is only as useful as the event and identity data your integration actually captures. The following is a practical application-level data checklist; it is not a claim that every field is natively returned by the controller.

Station	station_id, name, address/branch code, timezone, assigned_station_admin_id
Device	device_id, station_id, model, serial/label, connection status, IP reference, last_seen_at, firmware/version if exposed
Person	person_id, station_id, name, person category/type, status, created_at
Credential/Card	credential_id, person_id, masked_card_number, card technology, validity_start, validity_end, status
Access event	event_id, timestamp, device_id, credential/person reference if resolved, result, event code, reason if available, door/reader reference
Schedule	schedule_id, name, days, time intervals, holiday rules, assigned door/time section, sync state
Alert	alert_id, timestamp, device_id, event type, severity, status, acknowledged_by, resolved_at, resolution_note

13. Chart rules — avoid misleading analytics

• Total scans means authentication attempts, not unique visitors. If you need unique users, deduplicate by person/card and a defined time window.

• Granted and denied should be mutually exclusive outcomes for a single attempt. Define event mapping before charting.

• A door unlock relay command is not proof that the door physically opened. Use a door contact sensor for physical open/closed state.

• Device online status is not the same as door availability. Show them as separate signals.

• Do not label scan counts as washroom occupancy or people currently inside; the described setup is a card reader/door access controller, not an occupancy counter.

• Average unlock duration should only appear if the integration can measure a reliable unlock start/end or door-state interval.

14. Confirmed vs needs-validation feature matrix

Device registration in SmartPSS Lite	Confirmed in tested setup	Super Admin/device onboarding or controlled integration setup; station admin sees assigned device.
Online status	Observed in test	Show connection status.
Person management	Person record/profile displayed	Build People & Cards management.
IC card enrollment	13.56 MHz MIFARE card tested	Build card enrollment; confirm exact write/read workflow.
Card validity dates	Test dates were visible	Show validity start/end.
User type/status	Normal User / normal status shown	Expose supported categories/statuses only.
Schedules/door permissions	Fields present in device records	Build schedules/permission UI; verify write and enforcement.
Door unlock relay	Lock operation tested with ZKTeco LM-2802	Show unlock command/action result; do not infer physical door state.
Password / combined authentication	Documented capability	Do not include in station MVP until tested.
Anti-passback	Documented for original series	Optional/future; validate exact unit and reader configuration.

Door timeout/intrusion/duress/tamper alarms	Documented capabilities; may need sensors/config	Alerts UI can be designed, but enable only for verified event types.
Door sensor input	One input specified; sensor operation needs physical verification	Show door open/closed only after wiring and event validation.
External Wiegand/RS-485 reader	Interfaces documented; not confirmed in test	Do not promise until tested.

15. MVP implementation order

MVP 1 — Operational foundation	Overview; Access Logs; People & Cards; Device & Door basic online status	These map most directly to features confirmed in the tested setup.
MVP 2 — Access control management	Access Schedules; card validity; door/time-section permissions; sync feedback	Schedule fields were present, but enforcement and write behavior must be tested.
MVP 3 — Live operations	Live Monitoring; recent activity stream; alert/event page for verified event types	Requires event ingestion and reliable refresh/push behavior.
MVP 4 — Analytics and refinement	Reports; richer trend charts; exports; notification preferences	Build after event data quality and event mappings are stable.

16. Suggested component counts by page

Overview	12
Live Monitoring	6
Access Logs	8
People & Cards	8
Access Schedules	7
Device & Door	7
Alerts & Events	6
Reports	7
Settings	6
Total	67 page-level components (including tables, filters, cards, charts and panels)

Counts are a planning inventory, not a requirement to render every widget simultaneously. Reuse shared components (date filters, status badges, data tables, detail drawers and empty states) across pages.

17. Final design recommendation

• Use a clean enterprise layout: fixed left sidebar, compact top bar, station context, date filter and consistent content grid.

• Use green for granted/healthy, red for denied/critical, amber for warning/pending and neutral grey for unknown. Never use color alone; include text labels and icons.

• Keep the Overview dense but readable: four KPIs, one primary trend chart, one outcome donut, one hourly bar chart, a device status panel and recent activity.

• Make every KPI/chart clickable into a pre-filtered detail page. This makes the dashboard operational rather than decorative.

• Include loading, empty, stale-data and connection-error states. A stale live feed must not look live.

• Log every admin action that changes a person, card, schedule, permission or device configuration.

Source basis and limitations

Primary source: the user-provided “DHI-ASI1201E — Feature list for SmartPSS Lite integration” (prepared 10 November 2026). It identifies tested functions separately from capabilities listed in Dahua documentation. The source specifically cautions that datasheet features are not automatically available in the installed SmartPSS Lite version and must be validated against the exact unit, firmware, wiring and client version.

This blueprint translates that feature list into a proposed station-admin web application. The page structure, component counts, analytics layout, data model and workflow recommendations are product-design proposals, not features already proven in the controller.

Model-specific notes from the supplied source: exact model is DHI-ASI1201E (not -D); the tested card was 13.56 MHz MIFARE; lock tested was ZKTeco LM-2802; datasheet record/card capacities vary by revision and should be checked before being stated in product UI or procurement material.

End of blueprint